LECTURE 5

LAST TIME

LECTURE 4: SUMMARY

Storage costs and convenience yields

Forwards over time

- Value of forwards change in t
- Forward prices change in T
- Forward prices change in t

Risk of change $\implies$ need to hedge

THIS TIME

LECTURE 5 OUTLINE

Forwards and futures

Application: MGRM's failure

WARM-UP

WARM-UP 1: DISCONNECTED F and p?

We found the formula $F_0(T) = (1+r)^T p_0 + c$ by LoOP

For what kinds of underlying might this formula not be right?

I.e. when is forward price disconnected from underlying

WARM-UP 2: HEDGING

You are playing roulette in Monaco

You put 10,000€ on red

A risky bet!

Or is it necessarily? How could this bet actually reduce your risk?

WARM-UP 3: SKETCH FORWARD CURVE

Suppose $r = 2\%$ and $p_0 = \$1$

Sketch forward curve $T \mapsto F_0(T) = p_0 e^{(r-y)T}$ for $y = -1\%, 2\%$, and 4%

Comment on what underlyings might correspond to the different curves

NORMAL BACKWARDATION & CONTANGO

Forward markets are in normal backwardation if $F_t(T) < \mathbb{E}[p_T]$

Expect to make money if long forward: $\mathbb{E}[p_T - F_t(T)] > 0$

Forward markets in contango if $F_t(T) > \mathbb{E}[p_T]$

Expect to make money if short forward: $\mathbb{E}[F_t(T) - p_T] > 0$

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NB: "Backwardation" used for "normal backwardation" or $F_t(T) < p_t$

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Can you say who pays the premium for insurance?

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Normal backwardation $\implies$ sellers/producers

Contango $\implies$ buyers/users

FORWARDS VS. FUTURES

FUTURES ARE ALMOST FORWARDS

Futures and forwards have (almost) the same economic function

But institutional setup is different

Futures are exchange-based

They are centrally cleared by a clearing house

The clearing house is also a central counterparty

Futures are marked-to-market—value of position always zero

Futures are settled daily or intra-daily

In contrast, forwards are bilateral and settled at maturity

FUTURES ARE STANDARDIZED

Futures are standardized on a number of dimensions

- Contract size (quanty)
- Delivery month (timing)
- Assets to be delivered (quality)
- Daily resettlement (anonymity/risk)
- Look up quotes: futures.tradingcharts.com/marketquotes/

Futures markets work only if enough volume for each contract

CLEARING AND SETTLEMENT

What does it mean for an asset (e.g. a stock) to change hands?

Assets (e.g. share certificates) used to be physically traded

- Inefficient with high trade volumes (why?)

Changed to record-based the London Stock Exchange

Keep leger and settle every two weeks, netted/cleared trades

- Find net positions counterparties owed, then physical settlement

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Ongoing debate: Continuous settlement for all markets?

CCPs AND CREDIT RISK

A futures is a contract with a central counterparty (CCP)

- CCP is long to every short, short to every long

If CCP credit worthy, other parties don't worry about default

But CCP better be credit worthy

- Needs to ensure other parties don't default too much

MARGIN REQUIREMENTS

Futures clearing houses hold margin accounts

They settle from counterparties' margin accounts

Thus, to trade futures, you must post initial margin

If the value of your contract decreases, post variation margin

INTEREST RATE RISK

Margining leads futures to come with interest rate risk

If underlying correlates positively with short rate

e.g. short Eurodollar/SOFR futures

i.e. agreed to borrow at fixed rate in the future

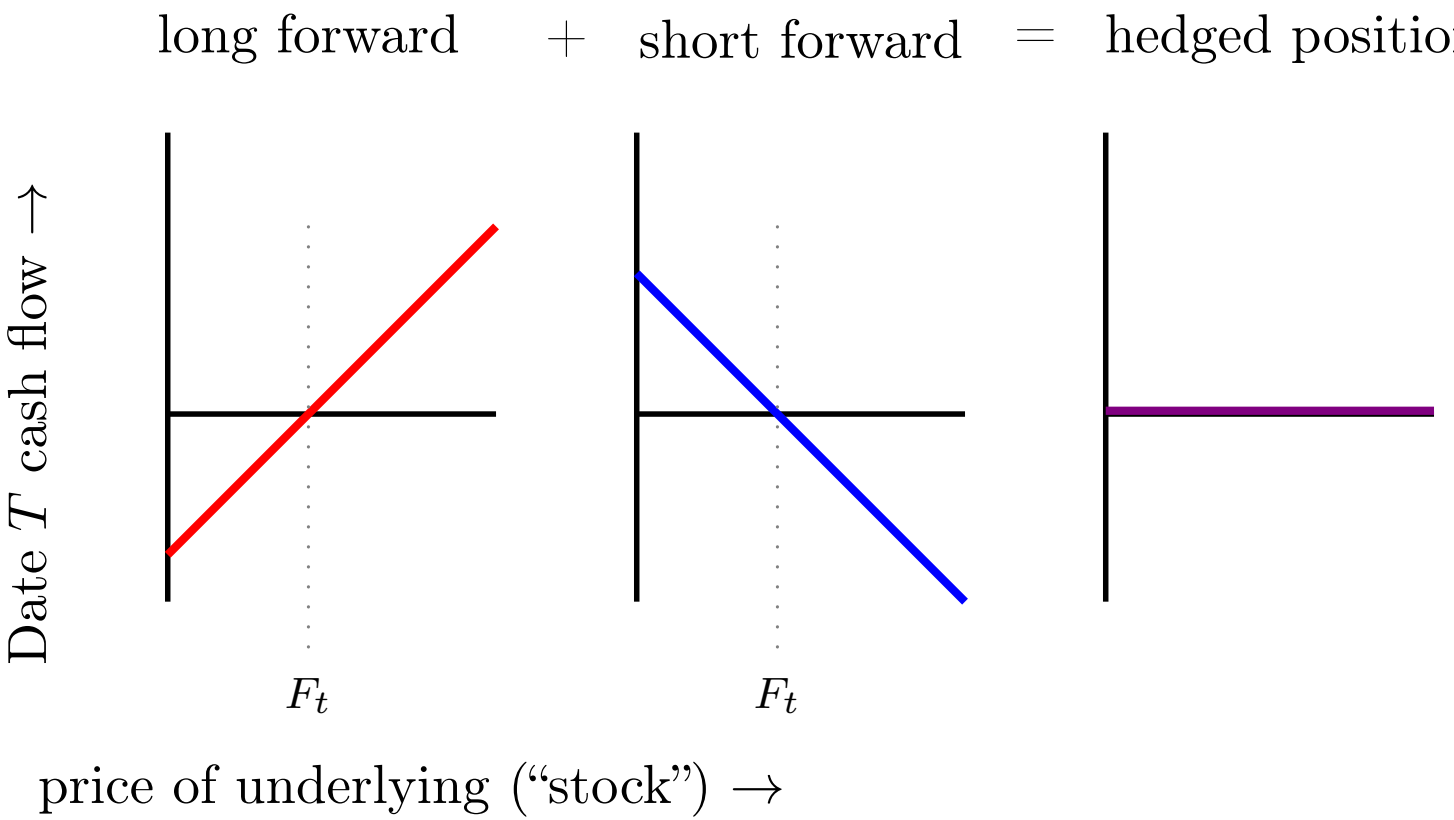
Then cash flows from margining can be reinvested profitably

Futures really have a second underlying: the short rate

Futures no longer linear but convex as function of short rate

HEDGING FORWARDS WITH FUTURES

RECALL BACK-TO-BACK HEDGE



HEDGING CONSIDERATIONS

PERFECT HEDGE?

No hedge is perfect (in reality)

Some hedges are better than others

What makes a good hedge?

WHAT CAN YOU MATCH?

Matching cash flows

Matching instruments

- Better to hedge forwards with forwards or futures?

Matching assets

- Sometimes the underlying is a special asset
- E.g. a specific set of mortgages
- Can you hedge crude oil with gasoline?
- How can you hedge inflation?

OTHER HEDGING CONCERNS

Basis risk

Anything that causes the position and hedge to diverge

E.g. interest rate risk between forwards and futures

Other example? Maturity, underlying,....

Rollover risk

Contango and backwardation

OTHER HEDGING CONCERNS

- Managing cash flows
  - Margin calls
- Funding risk
  - Raising capital can send a negative signal to the market
- Accounting treatment
  - Mark assets to market but (usually) not liabilities

METALLGESELLSCHAFT

METALLGESELLSCHAFT

Metallgesellschaft (MG) lost over two billion USD in 1993–1994

Losses resulted from its positions in forwards and futures

150 banks rescued MG at cost of 1.9 billion USD

- Saving 58000 employees from joblessness
- Perhaps even saving clearing association of futures exchange

MG'S HISTORY

Founded in Frankfurt in 1881

Made metal to fuel Second Industrial Revolution

Expanded to US to meet growing European demand in 1887

Suffered during wars, recovered with German growth in 1940s

Expanded into energy business in 1980s

Sold forward contracts to buyers, hedged with synthetic storage

MG'S ORIGINAL POSITION

MG wrote fixed-price forward supply contracts to customers

- Customers were retail fuel suppliers and big manufacturers
- Gas prices fell in 1993; customers locked in low prices
- MG allowed them to hedge by selling forward
- Also took a mark-up of a few dollars a barrel

Contracts left MG exposed to rising energy prices

MG'S HEDGE

Entered long futures and OTC swaps for 160 milli barrels of oil

Futures on NYMEX: long futures for 55 milli barrels of oil

Swaps OTC: receive floating, pay fixed for c. 100–110 milli barrels

For comparison: barrels estimated spilt in BP oil spill: 4.9 milli

DESCRIPTION OF HEDGE

MGRM buys one-month futures

Agrees to buy outstanding total delivery obligation

At the end of month, MGRM rolls the contract

Buys oil at futures price, delivers on forward, sells remainder

Buys one-month futures on remaining delivery obligation

IMPERFECT HEDGE

Suppose you are short a forward and want to hedge in the market

- You can use futures
- But liquid instruments offer only an approximate hedge
- Not quite the perfect hedge achieved with long forwards

How can you try to profit from your hedge?

ROLL GAINS

Don't match maturity perfectly, enter "stacked" position

Hedge short long-dated forward with long short-dated futures

- Roll over, buying the total underlying of forwards
- Buy at futures price, deliver on forward, sells remainder
- Enter into new short-dated futures underlying

COST OF ROLL-OVER

Call the futures price $\Phi_t(T)$ and the underlying price S

Say you roll over futures position every day/week/month

At date $t+1$ you book

$$\text{P\&L}_{t+1} = S_{t+1} - \Phi_t(t+1)$$

Then enter new futures contract at $t+1$: $\Phi_{t+1}(t+2)$

When is $\mathbb{E}_t\left[\text{P\&L}_{t+1}\right] > 0$?

STACKED POSITION P&L EXERCISE

At Date t, $S_t = 100$

At Date $t+1$, there are two equally likely outcomes

$$S_{t+1} = 101 \text{ and } S_{t+1} = 100$$

The futures price today is $\Phi_t(t+1) = 100.4$

Is the market in contango? Bormal backwardation?

What is $\mathbb{E}_t\left[\text{P\&L}_{t+1}\right]$?

STACKED POSITION P&L SOLUTION

First compute the expected value of S_{t+1}

$$\mathbb{E}_t\left[S_{t+1}\right] = \frac{1}{2} \times 101 + \frac{1}{2} \times 100 = 100.5$$

Thus $\mathbb{E}_t\left[S_{t+1}\right] > \Phi_t(t+1) = 100.4$

The market is in normal backwardation not contango

Normal backwardation and contango are mutually exclusive

But $\Phi_t(t+1) > S_t = 100$: Beware $\mathbb{E}_t\left[S_T\right] > F_t(T) \not\Rightarrow S_t > F_t(T)$

HEDGING OR SPECULATING?

The stacked position yields positive expected profits when

$$\mathbb{E}_t\left[\mathrm{P\&L}_{t+1}\right] > 0$$

or

$$\mathbb{E}_t\left[S_{t+1}\right] > \Phi_t(t+1)$$

or when markets are in normal backwardation

MG case: in early nineties oil markets in normal backwardation

Stacked long position yielded (expected) "roll gains" but also risk

MARGIN CALL/LIQUIDITY SQUEEZE

In 1993 energy markets were in contango almost all year

MG suffered rolling losses

Late in year, spot prices tumbled

Oil fell from $19 a barrel in June to $15 in December

MG needed to increase its margin account at NYMEX

Couldn't find the funding to meet margin calls

NYMEX then demanded super-margin

Liquidity squeeze forced MG to rely on rescue capital

RECAP

LECTURE 5: RECAP

Forward can be hedged back to back

But an "imperfect hedge" may actually be speculation

Speculation comes with risk

Margin calls and funding risk can lead to failure

Hedging concerns

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